

Amphiphilic Poly(SBMA-co-EGDEA) Coatings for Marine Antifouling: Insights into Design Criteria for Hydrophobic Monomers

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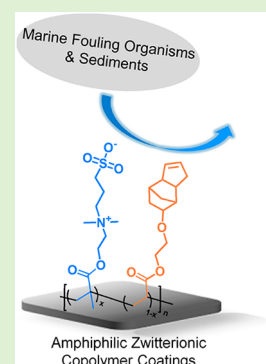


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ABSTRACT: Marine biofouling poses significant economic and environmental challenges, highlighting the need for effective antifouling coatings. We report amphiphilic poly(SBMA-co-EGDEA) copolymer coatings that resist both marine diatom adhesion and sediment adsorption. The coatings were synthesized via surface-initiated ATRP and RAFT polymerization using hydrophilic sulfobetaine methacrylate (SBMA) and hydrophobic ethylene glycol dicyclopentenyl ether acrylate (EGDEA). Interestingly, EGDEA exhibited a partial negative charge when coated on a surface, as confirmed by computational analyses. Since marine sediments are typically negatively charged and adhere strongly to hydrophilic surfaces, EGDEA's combination of hydrophobicity and negative charge effectively minimized sediment adsorption while maintaining resistance to diatom adhesion. Experimental results further showed that coatings with higher EGDEA content exhibited superior antifouling performance. These findings highlight the unique role of EGDEA in amphiphilic copolymers and provide valuable design insights for next-generation marine antifouling coatings.



1. INTRODUCTION

Marine biofouling, the accumulation of living organisms on artificial surfaces such as ship hulls and aquatic instruments, poses significant economic and environmental challenges by increasing fuel consumption and disrupting ecosystems.^{1,2} To address these issues, various strategies have been explored over the past few decades.^{1,3,4} Among them, tin-based paints were initially recognized as an effective solution for preventing marine fouling. However, due to their toxicity and severe environmental impact, the International Maritime Organization banned their use. Since then, the development of environmentally friendly antifouling strategies has gained significant attention.^{1,3–5} Antifouling and foul-release coatings are two primary approaches. Antifouling coatings rely on hydrophilic polymers to form hydration layers on surfaces, preventing the initial attachment of fouling organisms.³ In contrast, foul-release coatings use hydrophobic polymers to lower surface energy, weakening adhesion and allowing organisms to detach more easily under external forces such as ocean waves.⁶ Despite their advantages, both approaches have limitations: hydrophilic polymer coatings are prone to marine sediment adsorption, leading to functional degradation, while hydrophobic coatings are ineffective under static conditions without external forces to remove adhered organisms.^{7,8}

In this context, amphiphilic copolymers, which combine hydrophilic and hydrophobic components, have emerged as promising materials for marine antifouling coatings.^{9–21} For example, coatings composed of hydrophobic trifluoroethyl methacrylate and zwitterionic monomer have demonstrated excellent resistance to both marine diatoms and sediments.¹⁸

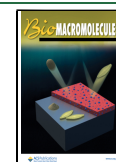
Due to the vast number of hydrophilic and hydrophobic monomer combinations that can be constructed, numerous amphiphilic copolymers have been synthesized and evaluated for their antifouling efficacy.^{9–18} In our study, we focused on sulfobetaine methacrylate (SBMA) and ethylene glycol dicyclopentenyl ether acrylate (EGDEA) as the hydrophilic and hydrophobic monomers, respectively. Carboxybetaine (CB) and sulfobetaine (SB) are common zwitterionic molecules. CB, which is pH-sensitive and can alter its structure, charge, and functionality, is mainly used in drug delivery.²² In contrast, SB has a very low pK_a and minimal pH reactivity, making it advantageous for developing stable antifouling materials that remain unaffected by pH variations.²³ Based on these characteristics, SBMA was chosen as the hydrophilic component for synthesizing amphiphilic copolymers. Furthermore, the strong binding affinity with water, which facilitates the formation of hydration layers along the polymer chains, effectively prevents biofouling.²⁴ While hydrophobic monomers are generally thought to impede hydration layer formation and consequently exacerbate biofouling, previous studies have demonstrated that poly(EGDEA) effectively suppresses bacterial adhesion on medical implants.^{25,26} This unique combination of hydrophobicity and biofouling resistance was the key rationale for selecting EGDEA as the

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hydrophobic monomer in the synthesis of the amphiphilic copolymer. Additionally, recent studies have reported that EGDEA, when copolymerized with carboxybetaine methacrylate (CBMA), enhances marine antifouling performance, further supporting its suitability for this application.^{12,13} Although amphiphilic copolymer coatings composed of CBMA and EGDEA have been successfully utilized, their detailed antifouling mechanism has not been thoroughly investigated. Additionally, the impact of substituting CBMA with SBMA, another representative zwitterion, remains unexplored. Furthermore, a systematic study to determine the optimal zwitterionic monomer-to-EGDEA ratio has yet to be conducted. Since the performance of copolymer coatings is closely linked to their composition, it is essential to investigate coatings with varying zwitterionic monomer-to-EGDEA ratios and evaluate their antifouling efficacy.

In this study, we found that the combination of SBMA and EGDEA exhibited excellent marine antifouling performance, with higher EGDEA content leading to improved performance. Using various analytical techniques, we identified a strong correlation between the surface charge of the copolymer coating and its antifouling performance.

2. EXPERIMENTAL SECTION

2.1. Materials. Tris(hydroxymethyl)aminomethane (Trizma base, 99%), tris(hydroxymethyl)aminomethane hydrochloride (Trizma HCl, 99%), dopamine hydrochloride (DA, 98%), zirconyl chloride octahydrate ($\text{ZrOCl}_2 \cdot 8\text{H}_2\text{O}$, 98%), ethylene glycol dicyclopentenyl ether acrylate (EGDEA), [2-(methacryloyloxy)ethyl]dimethyl-(3-sulfopropyl)ammonium hydroxide (SBMA, 95%), N,N,N',N'',N''' -pentamethyldiethylenetriamine (PMDETA, 99%), and 4-cyano-4-(phenyl-carbonothioylthio)pentanoic acid (CTPA) were purchased from Sigma-Aldrich. 2,2,2-trifluoroethanol (TFE, >99%), zirconium(IV) chloride (ZrCl_4), copper(I) bromide (CuBr , 98%), and copper(II) bromide (CuBr_2 , 99%) were procured from Alfa Aesar. Azobis(isobutyronitrile) (AIBN, 98%) and [11-[(2-bromo-2-methylpropanoyl)oxy]undecyl]phosphonic acid (initiator, >98%) were purchased from Junsei and TCI, respectively. The marine benthic diatom, *Amphora coffeaeformis*, was obtained from the Korea Marine Microalgae Culture Center (KMMCC).

2.2. Initiator Coating on Solid Substrates. The solid substrates (Si/SiO_2) were first cleaned with ethanol using sonication for 5 min and then dried with a nitrogen gas stream. Next, they were submerged in a DA solution (10 mM, Tris buffer, pH 8.4) for 3 h and rinsed with deionized (DI) water. The polydopamine (pDA)-coated substrates were then placed in an ethanol solution of ZrOCl_2 (5 mM) for 10 min, followed by rinsing with ethanol and DI water, and drying with nitrogen gas. Finally, the substrates were incubated in an ethanol solution of the initiator (1 mM) for 10 min, rinsed again with ethanol and DI water, and dried under a nitrogen gas stream.

2.3. Formation of Poly(SBMA-co-EGDEA)_x Brushes on Si/SiO_2 Substrates via SI-ATRP. Before use, the EGDEA monomer was purified by passing it through a basic alumina syringe to remove inhibitors. The monomers (SBMA and EGDEA) were dissolved in varying ratios in a solvent mixture (TFE/MeOH = 2:1), and 1.5 mL of the monomer solution was added to a glass vial containing the initiator-coated solid substrate (Table 1).²⁷ The solution was then degassed by bubbling nitrogen through it for 1 h. Next, a copper/ligand solution (1.5 mL) containing CuBr (14.98 mg), CuBr_2 (4.62 mg), and PMDETA (27.86 mg) were introduced into the vial for polymerization. This copper/ligand solution was also degassed by nitrogen bubbling for 1 h before use. Polymerization was carried out at 70 °C for 20 h, after which the solid substrates were washed with TFE and MeOH and dried using a nitrogen gas stream. When working with EGDEA, all vials were wrapped in foil to protect them from light, and the experiments were conducted with care.

Table 1. Final Concentrations of Monomers Used for Preparing Each Type of Polymer Brushes

polymer brush		SBMA conc. (mM)	EGDEA conc. (mM)
poly(SBMA)		1750	0
poly(SBMA _x -co-EGDEA _y), x:y	5:5	875	875
	3:7	525	1225
	1:9	175	1575
poly(EGDEA)		0	1750

2.4. Dynamic Marine Diatom Adhesion Test with Silt. A suspension of marine diatom with a chlorophyll content of 3 $\mu\text{g}/\text{mL}$ was prepared. Natural sand sediment (silt) collected from the test site at Port Boryeong, South Korea (36°20'15.1"N 126°31'57.5"E), was added to the suspension at a concentration of (37.5 g/L). A 13 mL portion of this mixture was then spread onto the solid substrates and incubated in a MaXshakeRP2 reciprocating shaker (Daihan Scientific Co.) at 20 °C and 50 rpm for 24 h. After incubation, the solid substrates were gently rinsed with artificial seawater. The diatom fouling behavior on the solid substrate was examined using fluorescence microscopy.

2.5. Synthesis of Poly(SBMA-co-EGDEA) via RAFT Polymerization. A solution containing EGDEA (2.40 g, 9.67 mmol), AIBN (0.0017 g, 0.011 mmol), CTPA (0.030 g, 0.11 mmol), and SBMA (0.30 g, 1.07 mmol) was prepared in DMSO (5 mL). This solution was transferred into ampules, degassed through three freeze–pump–thaw cycles, and then flame-sealed under vacuum. The ampule was heated at 70 °C for 72 h, then cooled to room temperature and opened to stop the polymerization process. The resulting product was precipitated in ethanol, vacuum-filtered, and dried under high vacuum. To further purify the polymer, it was dialyzed (molecular weight cutoff [MWCO] = 3500) against DI water for 3 d, followed by freeze-drying for 3–5 d to remove residual DMSO. Lastly, the product was stirred in ethanol for 3 d, vacuum-filtered, and dried under high vacuum to remove any remaining monomers in the copolymer.

2.6. RAFT-Synthesized Poly(SBMA-co-EGDEA) Coating on Solid Substrates. Before applying the poly(SBMA-co-EGDEA) coating, Si/SiO_2 and stainless steel substrates were cleaned with ethanol using sonication for 5 min and then dried with a nitrogen gas stream. The substrates were subsequently immersed in a DA solution (10 mM, Tris buffer, pH 8.4) for 3 h and rinsed with DI water. A solution of poly(SBMA-co-EGDEA) (10 mg/mL) was prepared in TFE and dropped onto the pDA-coated substrates, which were then spin-coated at 3000 rpm for 180 s. The polymer-coated substrates were immediately immersed in a methanol solution containing ZrCl_4 (0.1 M) for 10 min, followed by immersion in a TFE:ethanol (1:2) mixture containing ZrCl_4 (0.1 M) and then further immersed in TFE for 10 min. Finally, the substrates were rinsed with DI water and dried under a nitrogen gas stream.

2.7. Characterization of the RAFT-Synthesized Polymer. Molar mass of the RAFT-synthesized poly(SBMA-co-EGDEA) was determined by Tosoh EcoSEC HLC-8320 gel permeation chromatography (GPC) columns and a TSKgel guard column SuperAW-H. An HFIP solution of sodium trifluoroacetic acid (10 mM) was used as the eluent at a flow rate set to 0.3 mL/min at 40 °C. Calibration was performed using poly(methyl methacrylate) standards (160–675500 g/mol). The RAFT-synthesized poly(SBMA-co-EGDEA) was also analyzed by liquid chromatography–mass spectrometry (Agilent G6135BA, NFEC-2024-11-300749). Nuclear magnetic resonance (NMR) spectroscopy (AVANCE 400 MHz, Bruker, Billerica, MA) was used to determine the SBMA-to-EGDEA ratio in poly(SBMA-co-EGDEA), using deuterated TFE as the solvent.

2.8. Characterization of the Polymer-Coated Substrates. X-ray photoelectron spectroscopy (XPS) was performed using a Thermo Scientific NEXSA G2 spectrometer (Waltham, MA) with an Al K α X-ray source. The coating thickness on solid substrates was measured using an α -Step (P-7 Stylus Profiler, KLA-Tencor) and an

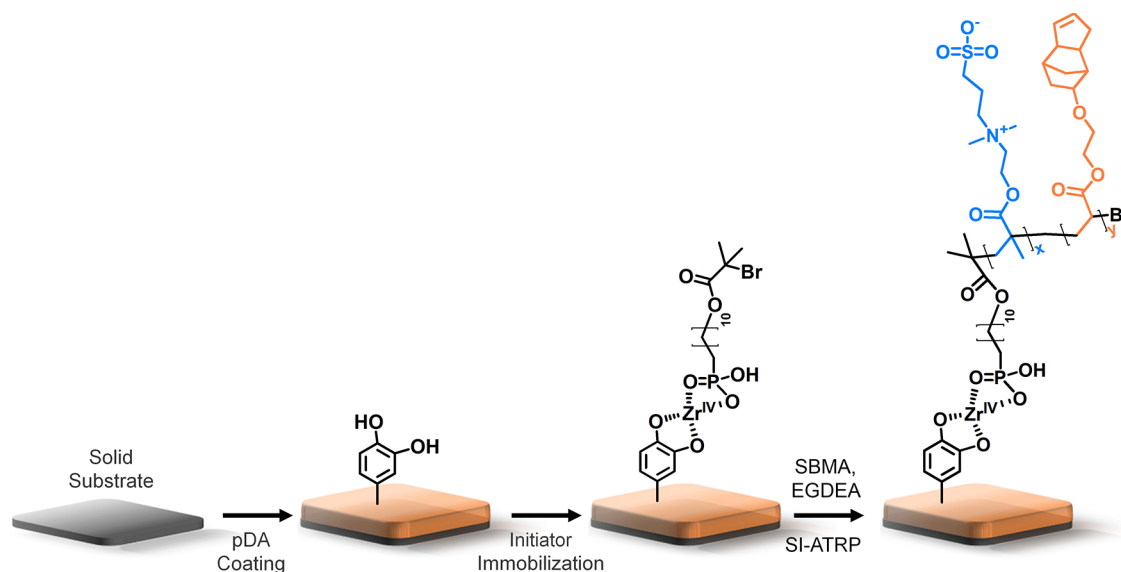


Figure 1. Schematic illustration for poly(SBMA_x-co-EGDEA_y) brush formation on solid substrates.

α -SE spectroscopic ellipsometer (J.A. Woollam) within a wavelength range of 390–900 nm at a 70° incidence angle. The data were analyzed using the Cauchy model in CompleteEASE software with a refractive index defined as $n(\lambda) = 1.45 + 0.01/\lambda^2$. Static water contact angle (WCA) measurements were conducted using a Phoenix-10 system (Surface Electro Optics, Suwon, South Korea) to evaluate the coatings' hydrophobicity. Atomic force microscopy (AFM) images were captured in noncontact mode with an XE-7 microscope (Park Systems, Suwon, South Korea). Fluorescence images were obtained using an optical microscope equipped with a fluorescence imaging system (LV100ND; Nikon, Tokyo, Japan). Fourier-transform infrared (FT-IR) spectra were obtained using a vacuum infrared spectrometer (Bruker, VERTEX 80v).

2.9. Statistical Analysis. Three independent samples were prepared. The data are presented as the mean $\pm$ standard deviation. Statistical analysis was carried out using an analysis of variance (ANOVA) followed by Tukey's Test for multiple comparisons (significance level $\alpha = 0.05$).

3. RESULTS AND DISCUSSION

The coating process started by immersing the substrate in a 10 mM dopamine (DA) solution prepared in Tris buffer (pH 8.5)

Table 2. Water Contact Angles and Coating Thicknesses of Various Polymer Brushes on Si/SiO₂ Substrates Prepared via SI-ATRP

polymer brush		WCA (°)	thickness (nm)
poly(SBMA)		<10	32.9 $\pm$ 5.0
poly(SBMA _x -co-EGDEA _y), x:y	5:5	55.9 $\pm$ 5.0	70.8 $\pm$ 10.2
	3:7	65.9 $\pm$ 5.4	124.4 $\pm$ 5.7
	1:9	82.4 $\pm$ 3.3	92.7 $\pm$ 4.8
poly(EGDEA)		89.9 $\pm$ 1.4	232.3 $\pm$ 31.9

for 3 h to introduce catechol groups onto the surface (Figure 1). A metal ion-mediated coordination strategy was then used to immobilize the ATRP initiator on the solid substrate, utilizing catechol groups as anchoring sites.²⁷ Among various metal ions considered, Zr(IV) was selected due to its ability to form stable coordination bonds with functional groups such as catechol, sulfonate, and phosphonate.²⁸ Unlike conventional initiator immobilization methods, our approach is compatible with a broad range of solid substrates due to the material-

independent coating capability of pDA. This versatility enables ATRP initiators to be anchored onto any solid surface, thereby facilitating surface-initiated atom transfer radical polymerization (SI-ATRP) on substrates of interest. Amphiphilic copolymer brushes were subsequently grown on the initiator-coated surface using SI-ATRP. To investigate the influence of EGDEA on the antifouling properties of poly(SBMA) brushes, a series of amphiphilic zwitterionic copolymer brushes composed of SBMA and EGDEA were synthesized by adjusting the monomer ratios. Polymer brushes with a higher EGDEA content than SBMA were prepared, with SBMA-to-EGDEA feed molar ratios of 5:5, 3:7, and 1:9. These polymer brushes were designated as poly(SBMA_x-co-EGDEA_y). Si/SiO₂ was chosen as the model substrate due to its suitability for a wide range of surface characterization techniques.

Water contact angle (WCA) measurements provided detailed insights into the surface properties of the prepared polymer brushes. The initial WCA of the uncoated Si/SiO₂ surface was 47.3 $\pm$ 5.1°. Following pDA coating and initiator immobilization, the WCA increased to 61.1 $\pm$ 7.8° and further to 80.7 $\pm$ 3.9°, indicating the successful immobilization of the hydrophobic initiator on the pDA-coated surface. Upon SI-ATRP, distinct changes in WCA were observed (Table 2). SI-ATRP of SBMA resulted in highly hydrophilic surfaces with WCA values below 10°, whereas the SI-ATRP of EGDEA produced hydrophobic surfaces with a WCA of 90°. For poly(SBMA_x-co-EGDEA_y) brushes with varying SBMA-to-EGDEA ratios, WCAs ranged from 56 to 82°, showing an increasing trend with higher EGDEA content. These results confirm the successful fabrication of poly(SBMA_x-co-EGDEA_y) brushes with tunable amphiphilicity via SI-ATRP. The thickness of the polymer brushes, measured using an α step surface profiler, was found to be greater than 30 nm for all samples. For further characterization of polymer brushes grown from the surface, an unbound sacrificial ATRP initiator (0.1 mM) was added to the reaction solution to obtain the free polymer chains. A previous study reported that the molecular weight of grafted polymer chains is comparable to that of free polymer.²⁹ The M_n and PDI of poly(SBMA) were 11,600 g/mol and 1.34, respectively, as determined by GPC in an aqueous sodium nitrate solution, calibrated with poly(ethylene

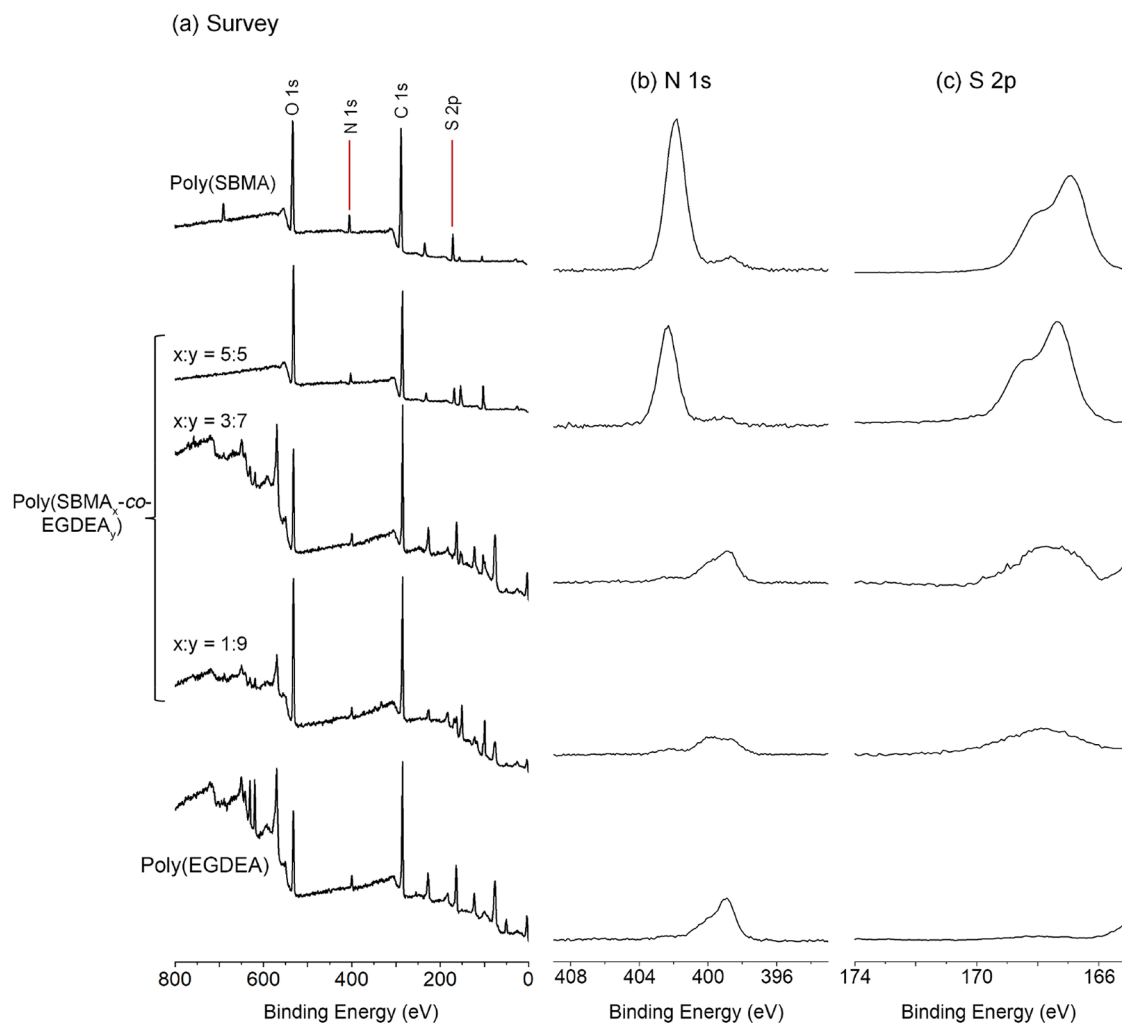


Figure 2. (a) Survey and (b and c) high-resolution (N 1s and S 2p) X-ray photoelectron spectra of Si/SiO₂ substrates coated with poly(SBMA), poly(SBMA_x-co-EGDEA_y), and poly(EGDEA).

glycol) standards (Figure S1). The grafting density of the poly(SBMA) brush was calculated to be 2.3 using the following equation:

$$\sigma = \frac{\rho \times h \times N_A}{M_n}$$

σ : grafting density

ρ : polymer density

h : polymer brush thickness

N_A : Avogadro's number

M_n : number-averaged molecular mass of the polymer chains

X-ray photoelectron spectroscopy (XPS) provided additional confirmation of polymer brush formation (Figure 2a). All poly(SBMA_x-co-EGDEA_y) brushes displayed N 1s (C–N⁺) and S 2p peaks (Figure 2b,c), corresponding to the quaternary ammonium and sulfonate groups of SBMA, respectively. High-resolution XPS spectra distinguished between the C–N⁺ peak at 401.6 eV, attributed to the quaternary ammonium group of SBMA, and the C–N peak at 399.7 eV, corresponding to the amino group of pDA (Figure 2b). The SBMA homopolymer exhibited the highest C–N⁺ to C–N ratio, which gradually decreased as the EGDEA content increased. This trend confirmed the successful synthesis of copolymer brushes with systematically varied SBMA and EGDEA compositions. FT-IR

spectroscopy further confirmed the presence of polymer brushes on the Si/SiO₂ surface. As shown in Figure S2, characteristic peaks corresponding to the SO₃[−] group (marked with red dots at 1033 and 1190 cm^{−1}) were mainly observed in the spectra of poly(SBMA) and poly(SBMA_x-co-EGDEA_y) brushes with compositions of $x:y = 5:5$ and $3:7$. In contrast, the poly(EGDEA) and poly(SBMA₁-co-EGDEA₉) brushes exhibited a distinct C–O stretching peak (marked with blue dots at 1107 cm^{−1}). Additionally, all polymer brushes displayed characteristic peaks associated with C–H stretching (~2950 cm^{−1}, marked with black dots) and C=O stretching (1730 cm^{−1}, marked with yellow dots).

The surface morphologies of the polymer-coated Si/SiO₂ substrates were analyzed using AFM. As shown in Figure S3, the poly(SBMA) brush exhibited a very smooth surface, with an RMS roughness of 2.1 nm, whereas the RMS roughness of poly(EGDEA) brush was 125.6 nm. Similar trends were observed for the copolymer-coated surfaces. The RMS roughness increased with higher EGDEA content in the copolymer: 15.0 nm for poly(SBMA₅-co-EGDEA₅), 19.9 nm for poly(SBMA₃-co-EGDEA₇), and 56.1 nm for poly(SBMA₁-co-EGDEA₉). The observed increase in surface roughness following the formation of amphiphilic copolymer brushes is likely attributable to microphase separation between the hydrophilic and hydrophobic components. These results

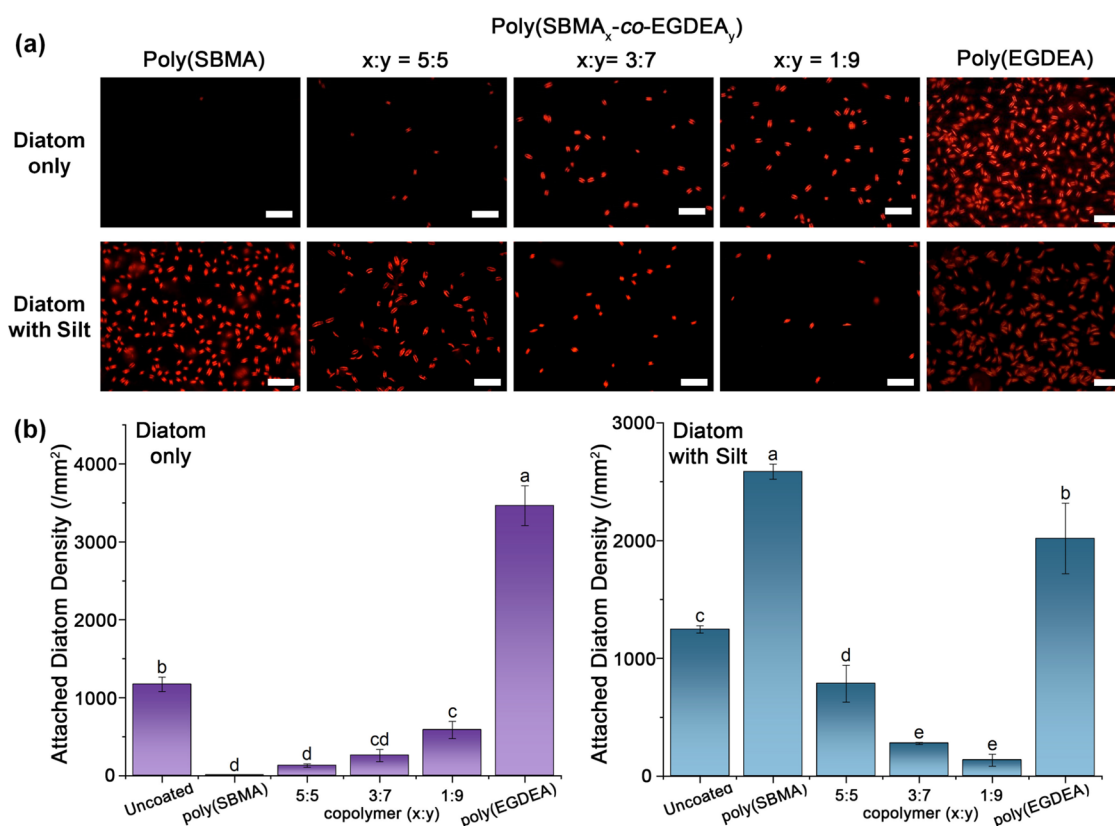


Figure 3. (a) Fluorescence images and (b) quantification of diatoms attached to the different coatings with and without marine sediment, silt. Scale bars represent 50 μm.

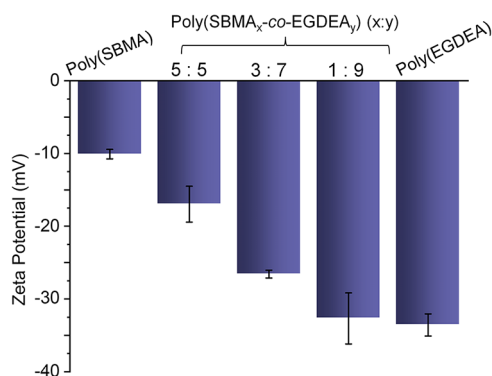


Figure 4. Surface zeta potential of poly(SBMA), poly(SBMA_x-co-EGDEA_y), and poly(EGDEA) brushes on Si/SiO₂.

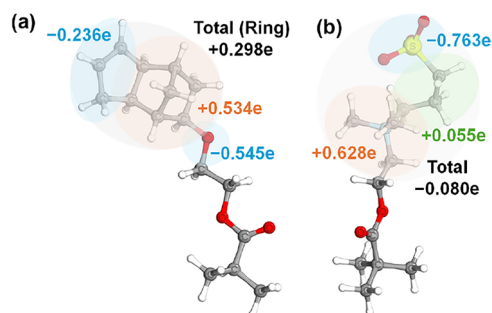


Figure 5. Merz-Kollman partial charges and the optimized molecular geometry of (a) EGDEA and (b) SBMA computed at the wB97X-D/6-31+G(d,p) level of theory.

indicate that both surface wettability and roughness can be tuned by adjusting the SBMA/EGDEA ratio in the copolymer brushes.

After characterizing the polymer brushes on solid substrates, a marine diatom adhesion test was conducted using two different conditions (Figure 3). In the first condition, the test was performed in the absence of marine sediments, silt. Uncoated Si/SiO₂ and Si/SiO₂ coated with various polymer brushes were incubated in a marine diatom suspension for 24 h (Figure 3a, top; Figure 3b, left). The results showed significant variation in attached diatom density depending on the surface characteristics of the polymer brushes. Poly(SBMA) completely suppressed diatom adhesion due to its strong hydrophilic properties.³⁰ In contrast, copolymer brushes exhibited increasing diatom attachment as the proportion of hydrophobic EGDEA increased. Notably, poly(EGDEA) strongly promoted diatom adhesion, with an approximately 3-fold increase in attached diatom density compared to uncoated Si/SiO₂ surfaces. These observations highlight the critical role of SBMA in inhibiting diatom adhesion under sediment-free conditions. In such environments, marine antifouling performance is primarily governed by the formation of a hydration layer, which acts as a physical barrier to prevent diatom attachment.^{31,32} Since SBMA is known to strongly promote hydration layer formation,²⁸ it plays a dominant role in antifouling performance. As the SBMA proportion decreases in copolymer brushes, hydrophobicity increases, which inhibits hydration layer formation, thereby reducing antifouling efficiency.

The marine antifouling properties of the prepared polymer brushes were also evaluated in the presence of marine

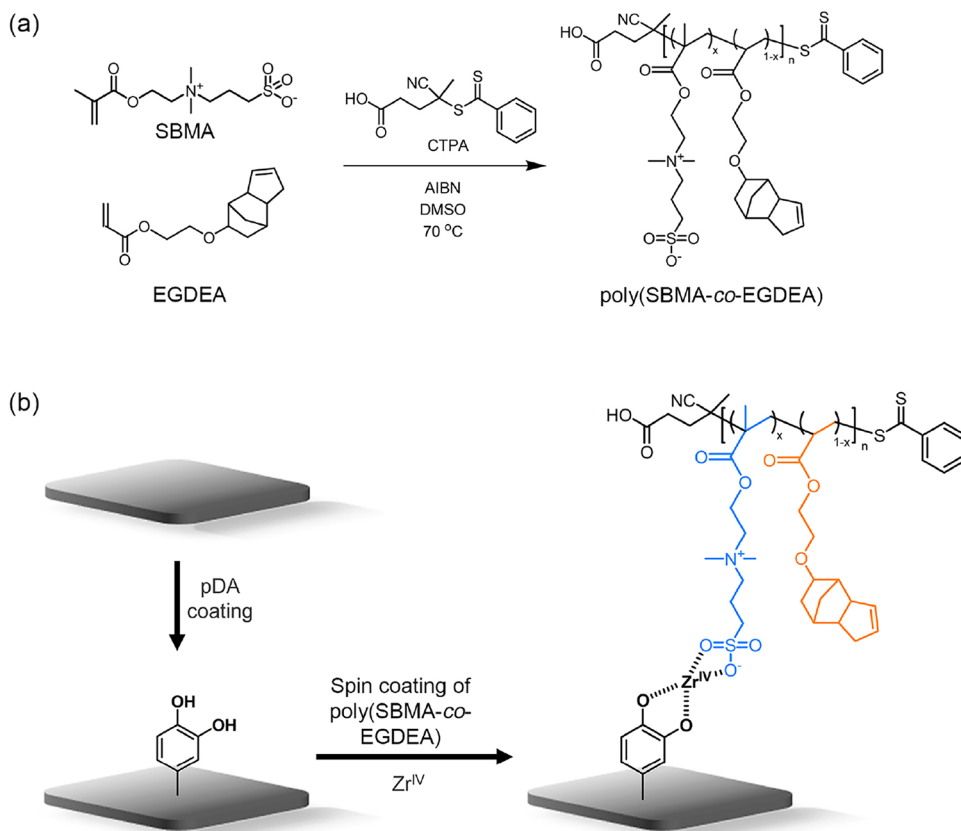


Figure 6. (a) A schematic route to the synthesis of poly(SBMA-co-EGDEA) via RAFT polymerization. (b) Schematic representation of the RAFT-synthesized poly(SBMA-co-EGDEA) coating.

sediments (Figure 3a, bottom; Figure 3b, right). Interestingly, the poly(SBMA) brush exhibited severe marine diatom adhesion. While diatoms also attached to uncoated and poly(EGDEA)-coated Si/SiO₂ surfaces, the highest diatom density was observed on the poly(SBMA) brush. To determine the underlying cause of this adhesion, we conducted additional adsorption experiments using silt alone. The results revealed substantial silt adsorption on the poly(SBMA) and poly(SBMA₅-co-EGDEA₅) brushes, whereas the poly(SBMA₁-co-EGDEA₉) and poly(EGDEA) brushes exhibited strong resistance to silt adsorption (Figure S4). This suggests that silt adsorption is responsible for the increased diatom adhesion on poly(SBMA), while the diatom attachment on poly(EGDEA) is primarily due to its hydrophobic nature. The adverse effect of SBMA on marine antifouling performance was also observed in poly(SBMA_x-co-EGDEA_y) copolymer brushes with varying SBMA-to-EGDEA ratios. As the SBMA content increased, diatom adhesion also increased due to SBMA's silt-adsorbing properties (Figure 3b, right). Among the copolymer brushes, poly(SBMA₁-co-EGDEA₉) exhibited the most effective suppression of diatom adhesion. These findings demonstrate how polymer chemical composition and surface properties influence marine biofouling, underscoring the importance of amphiphilic copolymers in achieving balanced marine antifouling performance.

To better understand the excellent marine antifouling properties of the poly(SBMA₁-co-EGDEA₉) brush, we further analyzed the surface charge of the polymer brushes. As shown in Figure 4, the poly(EGDEA) brush exhibited a highly negative surface charge (−33.6 mV), whereas the poly(SBMA)

brush showed a zeta potential of −17.0 mV. Due to the negative charge of poly(EGDEA), the zeta potential of the copolymer brushes decreased as the proportion of EGDEA increased. The negative charge of poly(EGDEA) has been previously reported by Williams et al., who prepared poly(EGDEA) coatings via radical polymerization and found that negative surface charge played a crucial role in preventing bacterial biofilm formation.³³ To further verify the negative charge of EGDEA, we conducted computational analysis. The molecular geometries of EGDEA and SBMA were optimized at the wB97X-D/6-31+G(d,p) level of theory,³⁴ and the partial charges were computed with the Merz–Kollman scheme.³⁵ All quantum chemistry calculations were performed with the program package Q-Chem 6.1,³⁶ revealing that the terminal part of EGDEA exhibited a significant negative charge of −0.24 (Figure 5a). Although the overall charge of the dicyclic ring in EGDEA is nearly neutral, we believe that its rigid structure leads to preferential exposure of the negatively charged terminal portion at the surface, resulting in a negative surface charge for the EGDEA coating. In contrast, while the terminal sulfonate group in SBMA also carries a strong negative charge of −0.763 (Figure 5b), the flexible nature of the SBMA structure prevents this group from being predominantly exposed at the surface, leading to an overall neutral average surface charge. In this context, the negative surface charge of amphiphilic poly(SBMA_x-co-EGDEA_y) brushes is attributed to EGDEA and is considered a key factor in achieving excellent marine antifouling performance in the presence of silt. Since silt particles carry a negative charge,³⁷ the combination of the copolymer's negative charge and hydrophobicity helps

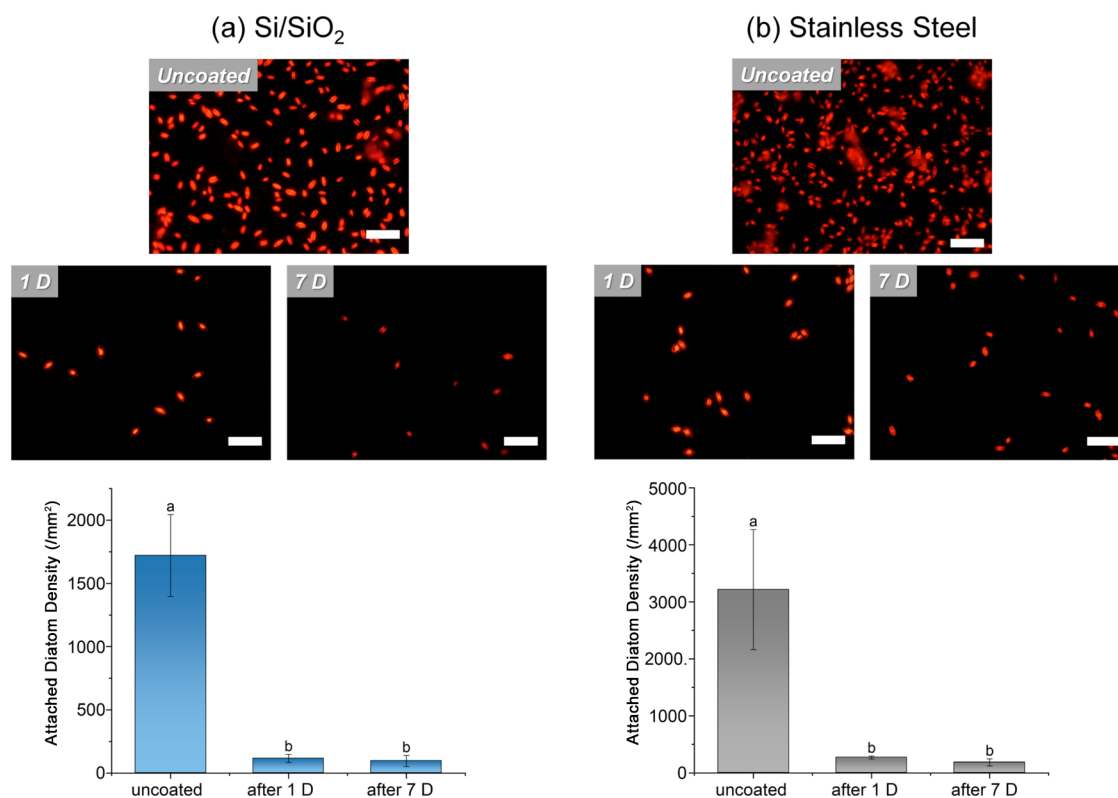


Figure 7. Fluorescence images and quantification of diatoms attached to the (a) Si/SiO₂ and (b) stainless steel substrates before and after the RAFT-synthesized poly(SBMA-co-EGDEA) coating. Diatom adhesion was analyzed after up to 7 days of incubation in the diatom suspension containing sediment.

minimize interactions with silt, thereby inhibiting diatom adhesion. Consequently, poly(SBMA₁-co-EGDEA₉), which exhibited the highest negative surface charge, demonstrated the most effective marine antifouling performance.

The polymer coating prepared via SI-ATRP has several limitations that hinder thorough solution-based characterization of the resulting polymers. Determining the copolymer's molecular weight and monomer composition is challenging, and applying this method for large-area copolymer coatings is practically unfeasible. Therefore, well-defined copolymers should be synthesized separately with controlled molar mass and composition before grafting. In this study, the copolymer was additionally synthesized via RAFT polymerization with an SBMA/EGDEA feed ratio of 1:9, as this composition exhibited the best marine antifouling performance (Figure 6a). RAFT polymerization was chosen because it is generally more suitable for industrial applications than ATRP, due to its simpler process, absence of metal catalysts, and greater environmental compatibility—particularly in applications requiring biocompatibility or large-scale production. After 3 days of RAFT polymerization, the molecular weight and polydispersity of the resulting poly(SBMA-co-EGDEA) copolymer were determined by GPC (Figure S5). The M_n and M_w of the copolymer were 35.2 and 61.3 kg mol⁻¹, respectively, yielding a molecular weight distribution ($\bar{D}$) of 1.74. The SBMA molar fraction in the copolymer was estimated using the integral ratio of the peak at 3.8–4.4 ppm (COOCH₂CH₂O– and –N⁺CH₂CH₂O–) to the peak at 5.3–5.7 ppm (–CH = CH–), revealing an SBMA-to-EGDEA ratio of 1:6.3 (Figure S6). After characterizing poly(SBMA-co-EGDEA), surface coating was performed. Catechol groups were introduced

onto the Si/SiO₂ solid substrate via pDA coating, and the copolymer was grafted onto the pDA-coated surface (Figure 6b).³⁰ Thickness measurements using an α -step surface profiler showed an average coating thickness of 98.42 nm. This relatively high coating thickness is attributed to Zr(IV)-mediated multiple cross-linking reactions. There are two possible cross-linking pathways: (1) between the catechol groups of pDA and the sulfonate groups in the copolymers, and (2) between sulfonate groups on different copolymer chains. These interactions resulted in the formation of a multilayered copolymer coating on the pDA-coated surface. The surface charge of the poly(SBMA-co-EGDEA) coating was measured to be –29.3 mV, which is comparable to that of the poly(SBMA₁-co-EGDEA₉) brush. Changes in WCA further confirmed successful copolymer grafting onto solid substrates. Due to the pDA coating,^{38–40} the copolymer coating could also be applied to other solid substrates. In addition to Si/SiO₂, stainless steel was used as another model substrate, as stainless steel is a major component of marine vessels, making its surface coating practically relevant. Regardless of the substrate, WCA increased after copolymer grafting, indicating successful surface modification (Figure S7). The grafting of the copolymer onto Si/SiO₂ and stainless steel substrates was further confirmed by XPS (Figure S8). Characteristic N 1s and S 2p peaks, corresponding to the quaternary ammonium and sulfonate groups of SBMA, newly appeared after copolymer grafting, verifying the presence of the SBMA-containing copolymer on solid substrates.

The marine antifouling properties of the copolymer-grafted surface were evaluated using a diatom suspension containing marine sediment. To assess the effect of seawater exposure on

antifouling performance, solid substrates were immersed in the marine diatom suspension for different durations, and their antifouling efficacy was compared with that of uncoated solid substrates. Compared to the uncoated substrates, which exhibited severe adsorption, the copolymer-grafted surfaces demonstrated significant antifouling effects, reducing diatom adhesion by 93% on Si/SiO₂ (Figure 7a) and 87% on stainless steel substrates (Figure 7b). Furthermore, the antifouling performance remained stable even after 7 days of incubation in a marine diatom suspension with silt. These results clearly indicate that (i) the copolymer grafting is robust and resistant to high-salt conditions, (ii) the EGDEA-containing amphiphilic copolymer effectively inhibits both diatom adhesion and silt adsorption, (iii) the combination with material-independent pDA precoating enables copolymer grafting onto stainless steel, making it more applicable for practical use.

4. CONCLUSIONS

In summary, we developed and evaluated amphiphilic poly(SBMA-co-EGDEA) copolymer coatings for marine antifouling applications. By optimizing the ratio of hydrophilic sulfobetaine methacrylate (SBMA) and hydrophobic ethylene glycol dicyclopentenyl ether acrylate (EGDEA), we achieved coatings that effectively resist both sediment adsorption and marine diatom adhesion. Among the tested compositions, the copolymer with an SBMA-to-EGDEA ratio of 1:9 exhibited superior antifouling performance due to its balanced surface properties, including controlled amphiphilicity and negative surface charge. The results demonstrated that poly(SBMA-co-EGDEA) coatings successfully mitigated biofouling even in the presence of marine sediments, overcoming the limitations of fully hydrophilic or hydrophobic coatings. Furthermore, coatings synthesized via RAFT polymerization and applied onto various solid substrates using polydopamine precoating exhibited excellent durability and stability in marine environments. Overall, this study highlights the potential of amphiphilic poly(SBMA-co-EGDEA) coatings as a promising and environmentally friendly alternative to conventional antifouling strategies.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.biomac.5c00875>.

GPC profile of the poly(SBMA) prepared in solution; FT-IR spectra of Si/SiO₂ substrates coated with poly(SBMA), poly(SBMA-co-EGDEA), and poly(EGDEA) brushes; AFM images of Si/SiO₂ coated with different polymer brushes; Optical images of different polymer coatings before and after silt adsorption test. The scale bars represent 100 μm; GPC profile of RAFT-synthesized poly(SBMA-co-EGDEA); NMR spectrum of RAFT-synthesized poly(SBMA-co-EGDEA); WCA images of Si/SiO₂ and stainless steel substrates coated with pDA and RAFT-synthesized poly(SBMA-co-EGDEA); Survey and high-resolution (N 1s and S 2p) X-ray photoelectron spectra of Si/SiO₂ and stainless steel substrates coated with pDA and RAFT-synthesized poly(SBMA-co-EGDEA) (PDF)

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Notes

The authors declare no competing financial interest.

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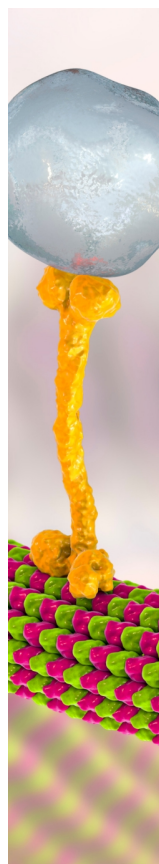
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